

Visium HD single cell spatial transcriptomics identifies heterogeneity of craniofacial muscles in human embryos



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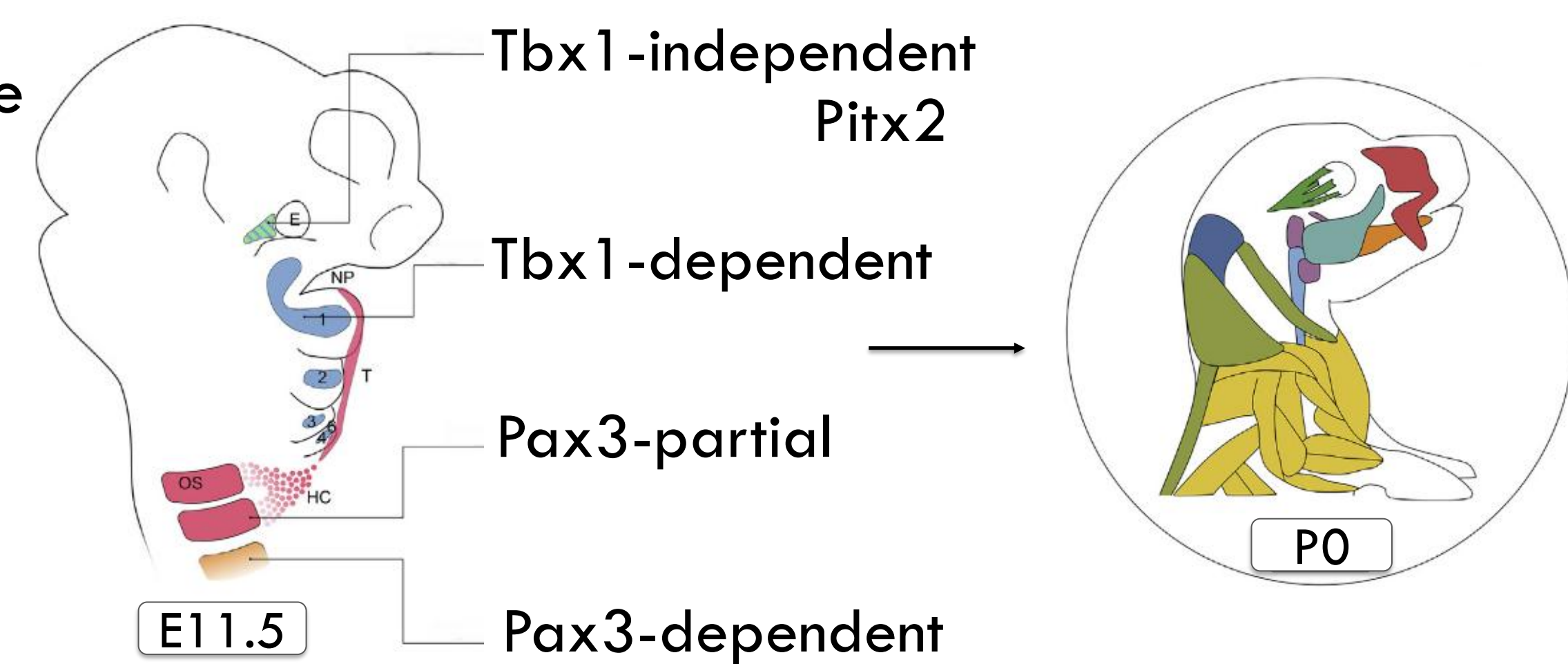
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- Craniofacial muscles are governed by distinct gene regulatory networks (GRNs) in mouse
 - Extraocular muscles (EOM) require the transcription factor Pitx2
 - EOM are independent of Tbx1

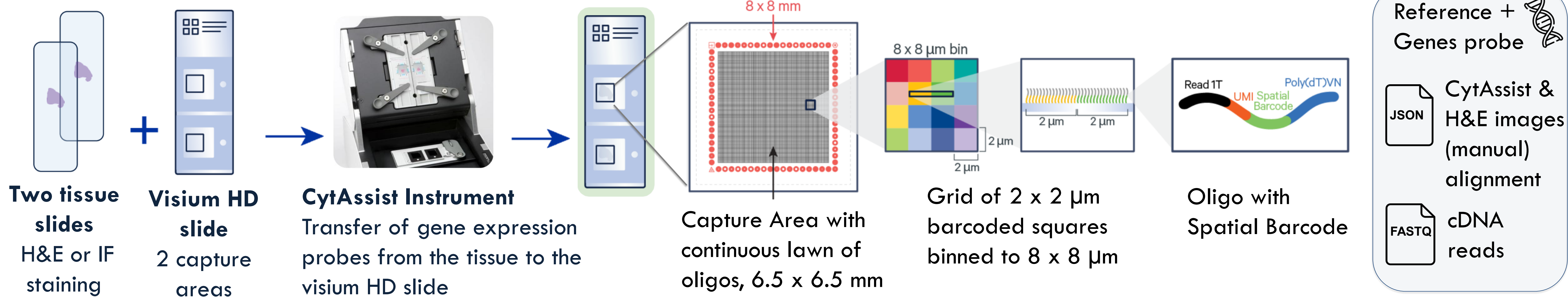
→ What about human?

- Spatial transcriptomics enables the study of gene expression within the context of tissue architecture
- Visium HD reaches single-cell scale resolution with continuous tissue coverage

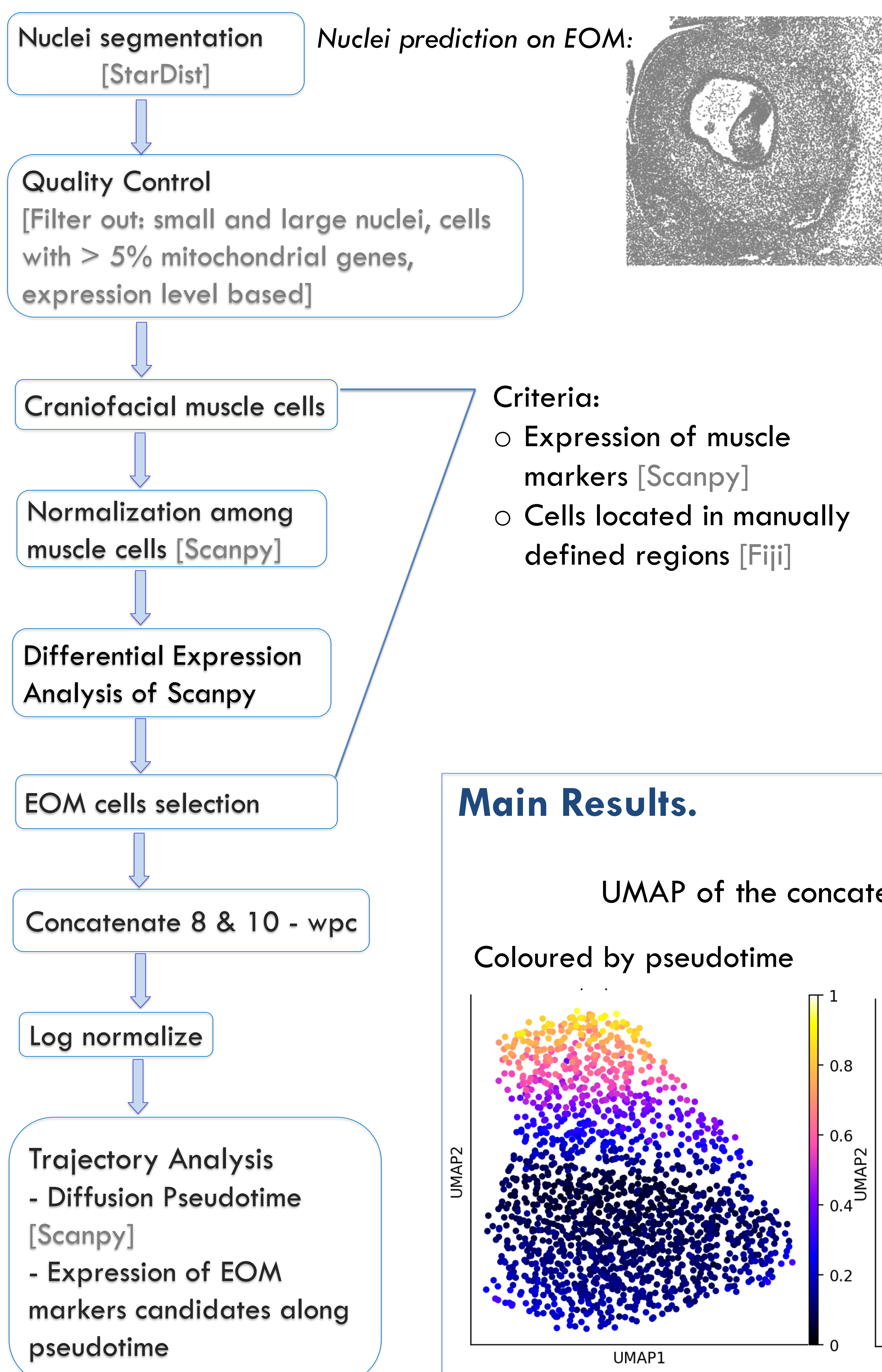


Grimaldi, A. (2021). *Current Opinion in Cell Biology*, 73.

Spatial Transcriptomics: Visium HD Technology



Computational Workflow



Embryo head sagittal sections 8 week post-conception (wpc)

H&E

CytAssist

Pitx2 on
8-wpc
section

EOM sections – 10 wpc

H&E

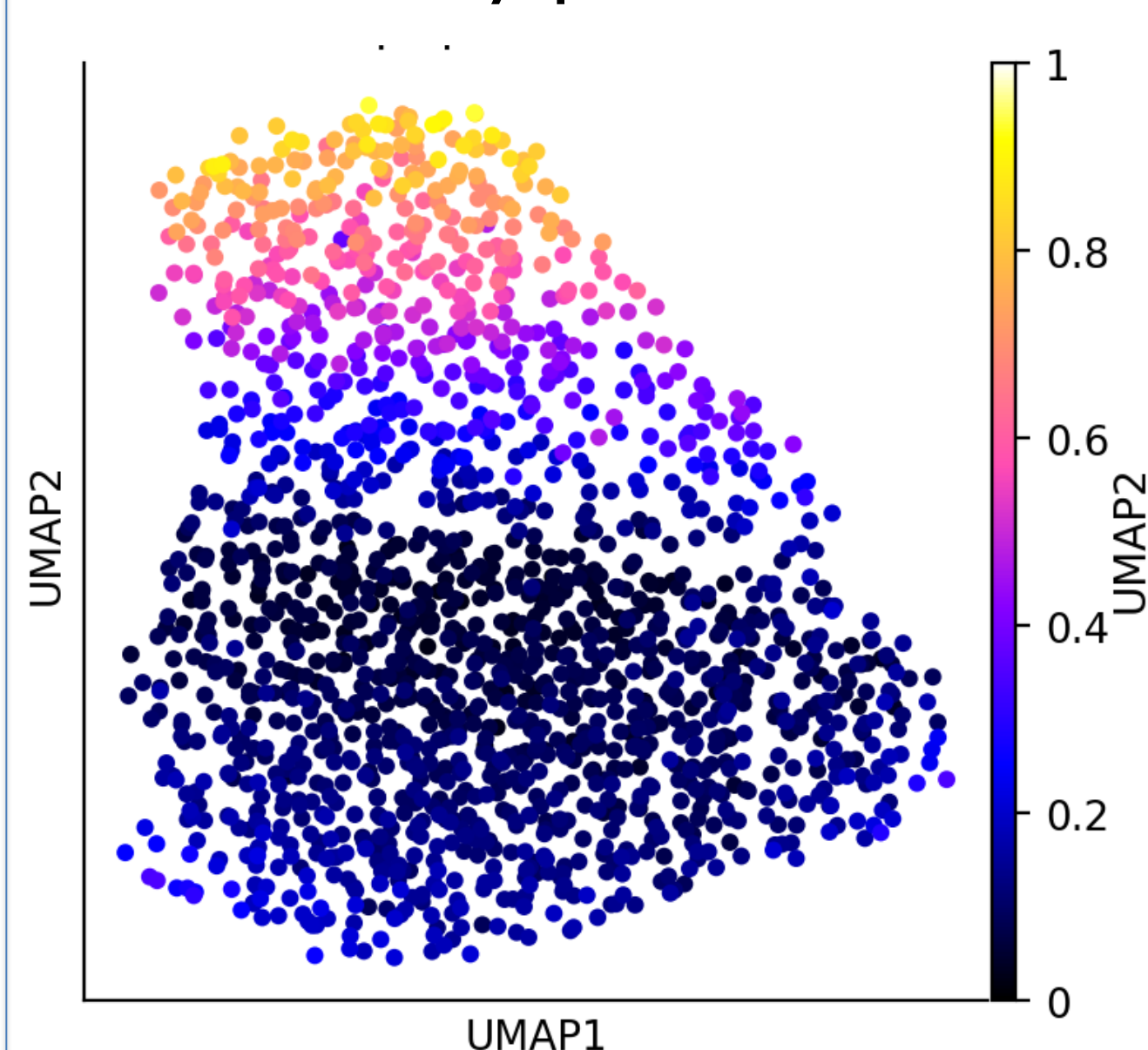
CytAssist

Myog on
8-wpc
section

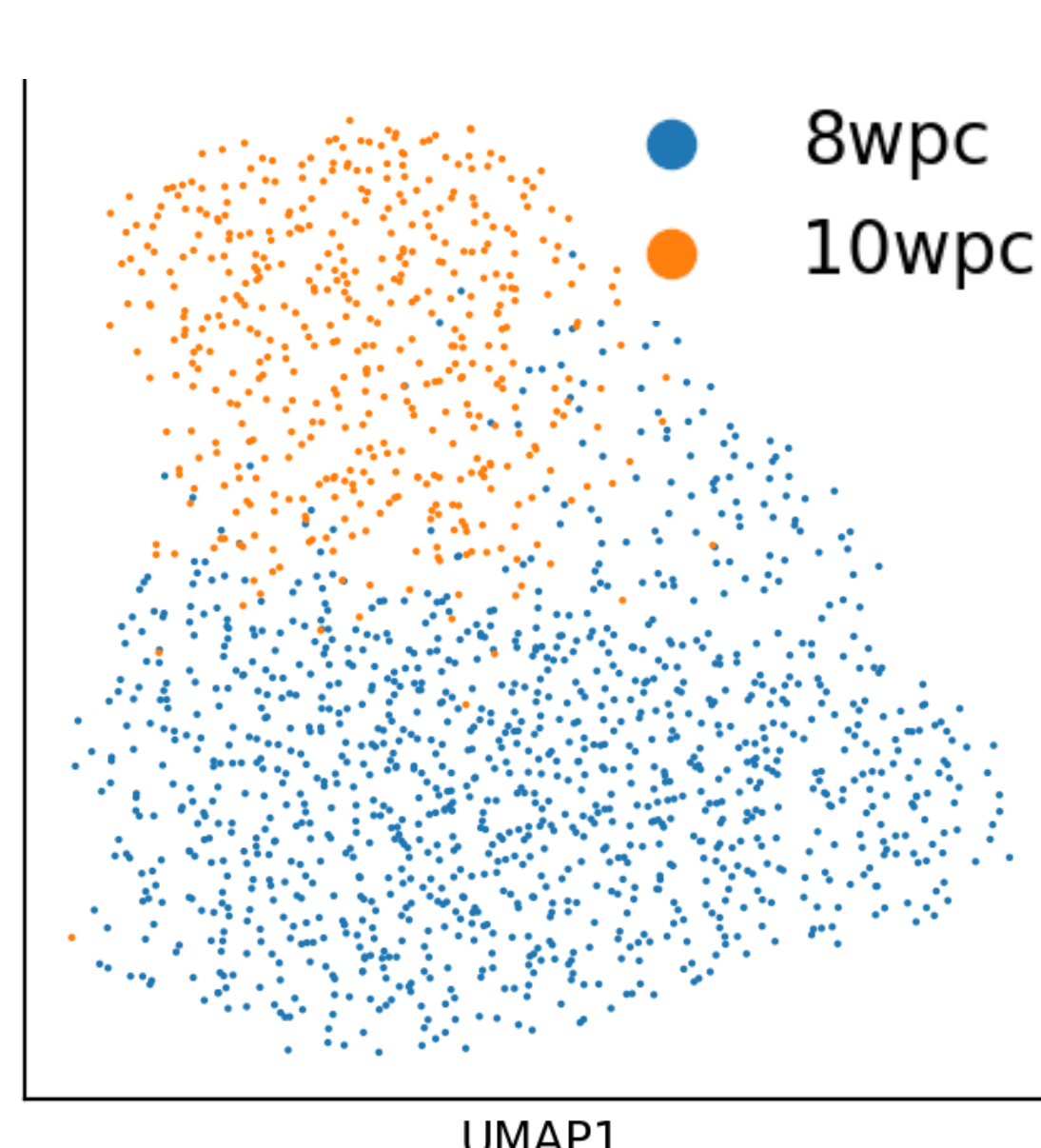
Main Results.

UMAP of the concatenated object

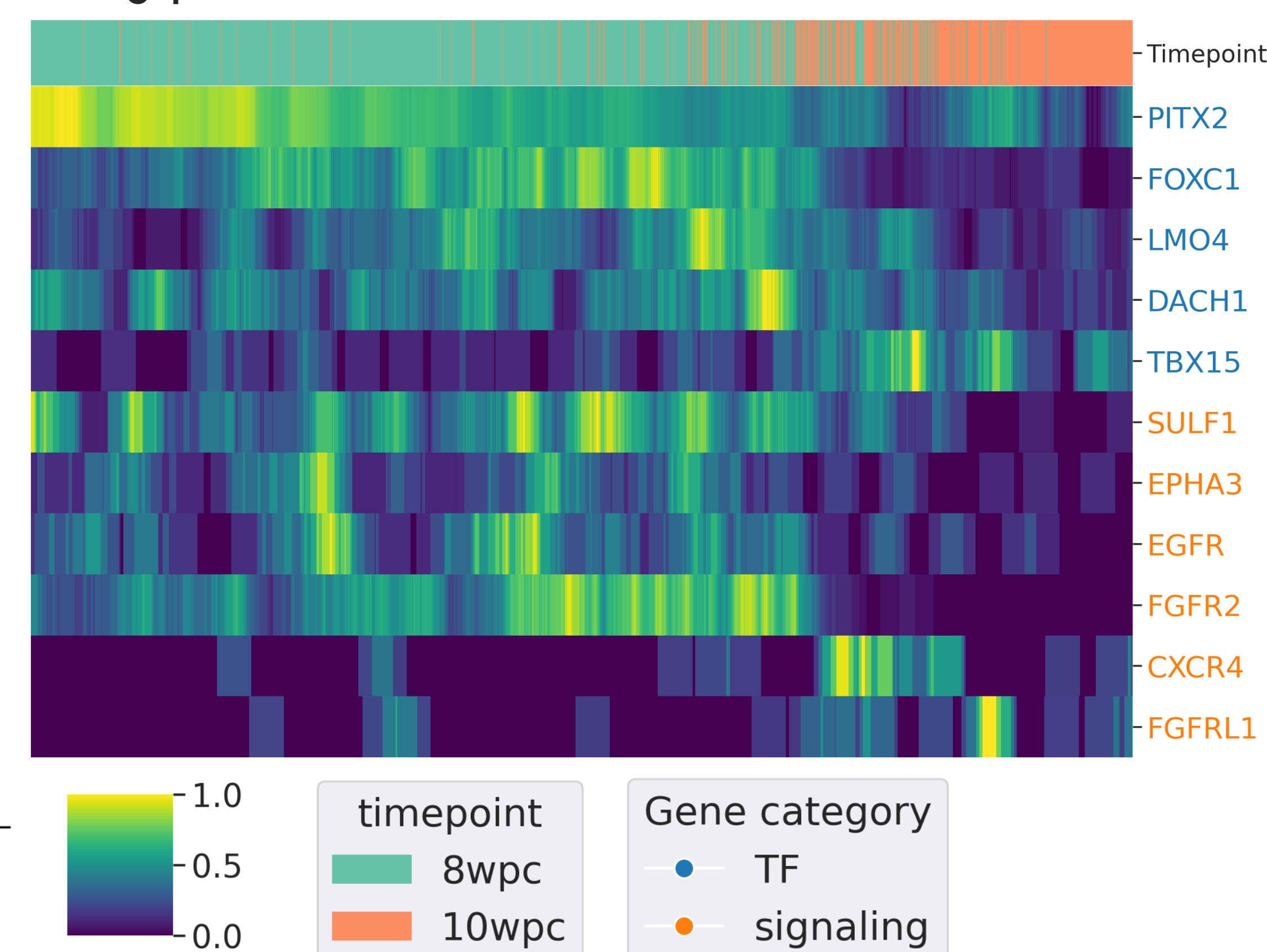
Coloured by pseudotime



Coloured by stage



Expression genes, upregulated in EOM, along pseudotime



Conclusion

- Spatial Transcriptomics of 8- and 10-wpc human embryo head sections was done
- A trajectory analysis was conducted
- Transcription Factors and signaling genes potentially activated during embryogenesis were identified